

FIDEL GONZALEZ VAZQUEZ

Montréal, QC, Canada

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Sommaire du profil

- Développeur de logiciels possédant 12 ans d'expérience dans le développement d'applications informatiques avec une spécialisation particulière en C/C++.
- Excellente connaissance de la programmation orientée objet
- Expérience de diverses méthodologies de développement de logiciels (SCRUM, RUP)
- Expérience de l'utilisation des technologies web (HTML5, JavaScript, Python, etc)
- Expérience dans l'enseignement supérieur
- Français (compétence professionnelle), anglais (compétence professionnelle), espagnol (courant)

Studies

Post-Graduate studies, Software Engineering	2015-2017
Université du Québec à Montréal, QC, Canada	
Post-Graduate studies, Applied Mathematics	2005
University UCLV, Santa Clara, Cuba	
Master's degree, Computer Engineering	2004
University UCLV, Santa Clara, Cuba	
B.Eng. Control Engineering	1998
University UCLV, Santa Clara, Cuba	

Technical Strengths

Programming :	C/C++, Python
Frameworks :	Asterisk, Qt
Embedded Systems :	Linux temps réel, POSIX, RTLinux, Partikle
Web Technologies:	HTML5, CSS, JavaScript
Database:	MySQL
Software:	Visual Studio, Visual Studio Code, Matlab, Latex, Squish
Tools:	GoogleTest, CMake, gcc, g++, gdb, subversion, git, PowerShell, Copilot, Conan

Professional Experience

Embedded software developer at OPAL-RT

September 2025- present

OPAL-RT delivers real-time and faster-than-real-time simulation platforms that empower engineers to test, validate, and refine complex control systems with confidence. From HIL, PHIL, RCP, and SIL to purely accelerated simulation, our scalable tools support every stage of development, from lab to production, driving innovation across industries with unmatched speed, accuracy, and flexibility.

Responsibilities:

- **API Development:** Collaborated in the design and implementation of new features for the RT-LAB API (C++/Python).
- **Software Quality & Debugging:** Participating in the maintenance cycle by investigating, diagnosing, and fixing software bugs, directly contributing to increased product stability and a more robust user experience.
- **Agile Collaboration:** Worked within an agile development team, participating in code reviews and sprint planning to deliver scalable tools that support the entire product development lifecycle, from lab to production.

Technologies, programming language, etc:

- C++03, Python, Matlab, Jira, Gitlab, real-time Linux, Java, Eclipse, Conan

Senior software developer at Toon Boom:

November 2024/ July 2025

Toon Boom Animation Inc. Is Canadian software company that specializes in the development and production of animation and storyboarding software for film, television, the Web, video games, mobile devices, training and education.

Responsibilities:

- **System Architecture:** Assisted in the analysis and design of system architecture to support scalable, high-performance animation tools.
- **Feature Development:** Designed, developed, and tested new software features (C++/Qt) from concept to deployment, enhancing the product for professional animators.
- **Software Maintenance:** Diagnosed and resolved known issues and bugs within existing applications, improving stability and user experience.

Technologies, programming language, etc:

- C++14, C++17, QT6, Bitbucket, Jira, Python

Senior software developer at Zimmer Biomet:

July/2017 – February/2024

Zimmer Biomet is a medical device company. The Montreal division designs innovative solutions that support orthopaedic surgeons and clinicians in alleviating pain and improving the quality of life for people around the world. The Montreal team mandates include the

development of 3D geometry algorithms, 3D modelling of anatomical data, programming of robotic movements and the design of user interfaces adapted to the medical environment.

Responsibilities:

- **Technical Leadership & Architecture:** Spearheaded the analysis and design of system architecture for software controlling robotic surgical planning and 3D anatomical modelling, ensuring reliability, safety, and scalability in a regulated (FDA) environment.
- **Full-Cycle Software Development:** Owned the end-to-end design, development, and testing of new software features, from initial concept through to validation, enhancing the surgeon's interface and robotic control algorithms.
- **Agile Project Management:** Served as Scrum Master, facilitating sprint planning, daily stand-ups, and retrospectives to drive team efficiency, improve workflow, and ensure on-time delivery of complex software milestones.
- **Team Development & Mentorship:** Acted as a primary mentor for interns and new team members, fostering a culture of technical excellence and accelerating their integration and productivity.
- **Quality Assurance & Compliance:** Oversaw comprehensive system testing and debugging processes to meet stringent medical device standards. Produced and maintained critical technical documentation and provided expert-level technical support.

Technologies, programming language, etc:

- C++14, C++17, Boost, QML, Qt5, Visual Studio, Visual Studio Code, GoogleTest, Python, PyCharm, JavaScript, Git, Subversion, Jira, mainly Windows environment, Squish, Cmake, Figma, Solidworks.

Software Developer Analyst at Aheeva:

July/2013 – July 2017

Aheeva is Canadian IT enterprise developing VoIP solutions for contact centers.

Responsibilities:

- **Architecture & Development:** Assisted in designing system architecture and developed new software components for call processing and contact centre logic.
- **Testing & Debugging:** Performed system testing and debugging to maintain platform reliability and performance.
- **Technical Ownership:** Created technical documentation and provided support, ensuring smooth deployments and system understanding.

Technologies, programming language, etc:

- C, multithreading, C++11, Asterisk, Jira, Linux /Windows, gcc, g++,gdb, Wireshark

Software consultant through the CEDAI company

2006-2010

CEDAI is Cuban national enterprise leader in control automation

solutions

Responsibilities:

- Design and development of the real-time applications
- Real-time system configuration
- Design, development and implementation of different applications related to the oil industry and the energy monitoring and control in hotels.

Technologies, programming language, etc:

- C, multithreading, C++98, RTLinux, Partikle, D-Bus, MySQL, Linux /Windows, Wireshark, gcc, g++,gdb, Qt4.